

Video Slides W2: Gains from Trade

ECO 101: Principles of Microeconomics

Prof. Freitas, University of Toronto

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Opportunity Cost with Production

Topic: Gains from Trade

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Video Learning Objectives

- Explain how to adapt the Opportunity Cost framework to a situation where we have to allocate resources to the production of different goods.

Question

I Want to start an Etsy business, what should I produce?

- T-shirts?
- Cute wooden tables?

Don't forget...

1. Costs = Opportunity Costs.
 - What resources so I use?
 - What are the alternative uses of my resources?
- 2 Simplify complex reality by making **assumptions**.

Example: Carpenter or Chef?

Simplifying assumptions.

- Resource: Everyone works for 10 hours a day.
- 2 goods: tables or meals.
- Production capabilities.
 - Ann: 3 tables an hour or 3 meals per hour.
 - Bill: 1 tables an hour or 2 meals per hour.

(Opp.) Cost of Production

What is the cost of producing a table or a meal given the number of units produced in an hour?

- Ann: 3 tables an hour or 3 meals per hour.
- Bill: 1 table an hour or 2 meals per hour.

	OC_{TABLE}	OC_{MEAL}
Ann		
Bill		

Cost of Production

What is the cost of producing a table or a meal given the number of hours needs to produce 1 unit?

- Ann: $\frac{1}{3}$ hour for 1 table or $\frac{1}{3}$ hour for 1 meal.
- Bill: 1 hour for 1 table and $\frac{1}{2}$ hour for 1 meal.

	OC_{TABLE}	OC_{MEAL}
Ann		
Bill		

OC captures Trade-offs

OC: Opportunity Costs

- If you want to produce one more **table**, how many fewer **meals** do you produce?
- If you want to produce one more **meal**, how many fewer **tables** do you produce?

Note:

- Trade-offs expressed in **units** of goods.
- Implicitly assuming that you're **using all resources**.

Absolute vs. Comparative Advantage

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Video Learning Objectives

- Define and determine Absolute and Comparative Advantage.

Two Questions

Why do we need cost information?

- Who is better (overall) at producing meals? At tables? Both?
- Who should specialize in meal production? .

Two Questions

- Who is better (overall) at producing meals? At tables? Both?
 - Absolute Advantage.
- Who should specialize in meal production?
 - Comparative Advantage.

Absolute Advantage

Definition: An individual person/firm/country has an absolute advantage in a good/activity if they

- Can produce more output with same inputs
- $\iff$ Can produce same output with fewer inputs

Example: Absolute Advantage

Ann: 3 tables an hour or 3 meals per hour.

Bill: 1 table an hour or 2 meals per hour.

- With one hour what can each produce?

	Tables	Meals
Ann	3 tables	3 meals
Bill	1 table	2 tables

Example: Absolute Advantage

Ann: 3 tables an hour or 3 meals per hour.

Bill: 1 table an hour or 2 meals per hour.

- How many hours to produce 1 extra good?

	Tables	Meals
Ann	$\frac{1}{3}$ hr	$\frac{1}{3}$ hr
Bill	1 hr	$\frac{1}{2}$ hr

- Does this mean that Ann should produce everything?

Comparative Advantage (CA)

Definition: Individual person/firm/country has a comparative advantage in a good/activity if the **opportunity cost** (OC) of producing the good is lower for that individual **compared** to others.

Comparative Advantage (CA)

OC: Opportunity Costs

	OC_{TABLE}	OC_{MEAL}
Ann	1 meal	1 table
Bill	2 meals	0.5 tables

- No difference in OC $\Rightarrow$ no comparative advantage.
- If Ann has a CA in good X then Bill has the CA in good Y

Summary

- Comparative advantage determined by lowest OC. Specialization determined by comparative advantage.
- Absolute advantage determined by who is most productive at producing a certain good.

Trade and Exchange Rates

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Video Learning Objectives

- Explain why a difference in Opportunity Costs leads to gains from trade
- Calculate terms of the trade.

Trade

What if Ann and Bill had to figure out trading by themselves.

- Who should produce what good?
- If they decide to (partially) specialize, what should the exchange rate be?

Who (Partially) Specializes in Meals?

OC: Opportunity Cost

- The person with the lowest Opportunity Cost in meals should produce it first.

	OC_{TABLE}	OC_{MEAL}
Ann	1 meal	1 table
Bill	2 meals	0.5 tables

- Why Comparative advantage and not Absolute Advantage?

Example: Ann and Bill Trade

Think barter: no cash, exchange rate in terms of goods.

Bill proposes to give Ann 3 meals in return for 2 tables. Does Ann accept?

i.e Ann produces 2 more tables and exchanges that for 3 meals

ANN	Yes, TRADE	NO trade
Tables produced	15+2	15
Tables consumed	15	15
Meals produced		15
Meals consumed		15

Example: Ann and Bill Trade

Think barter: no cash, exchange rate in terms of goods.

Ann proposes to give Bill 2 tables in return for 3 meals. Should Bill accept?

i.e Bill produces 3 more meals and exchanges that for 2 tables.

BILL	Yes, TRADE	NO trade
Tables produced		5
Tables consumed		5
Meals produced	10+3	10
Meals consumed	10	10

Bill	1 table/hr or 2 meals/hr. Comparative advantage in meals.
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Trade

What if Ann and Bill had to figure out trading by themselves.

- Who should produce what good?
- If they decide to (partially) specialize, what should the exchange rate be?

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Trade

Trade is voluntary. They will only accept trade if it makes them better off than what they can do themselves.

- What is the maximum the buyer is willing to pay?
- What is the minimum the seller is willing to accept?

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Price of 1 Meal?

Think barter: no cash, exchange rate in terms of goods.

	OC_{TABLE}	OC_{MEAL}
Ann	1 meal	1 table
Bill	2 meals	0.5 tables

- ANN: The max. willing to pay for 1 meal is _____
- BILL: The min. willing to accept for 1 meal is _____

Exchange Rate

Think barter: no cash, exchange rate in terms of goods.

	OC_{TABLE}	OC_{MEAL}
Ann	1 meal	1 table
Bill	2 meals	0.5 tables

- The exchange rate (E) for 1 meal will lie between

$$OC_{MEAL}^{BILL} \leq E \leq OC_{MEAL}^{ANN}$$

Gains from Trade

- Differences in opportunity cost open up potential for gains from trade.
- Is trade always good?
- Given what people want, is there an efficient way to produce that combination?

Summary

OC: Opportunity Cost

- Trade moves production from high OC producer to low OC producer.
- Each of us moves towards (partial) specialization in the good we have a **comparative** advantage in.
- Trading prices are expressed units of a good and will be determined by opportunity costs.

Production Possibilities Frontiers

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Video Learning Objectives

- Draw (individual and joint) Production Possibility Frontiers (PPFs)
- Show graphically using PPFs why trade makes people better off.
- Relationship between the shape of the PPF and Opportunity Cost.

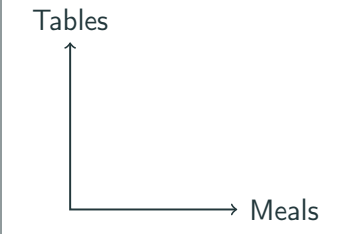
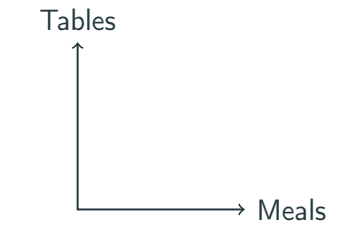
Definitions

PPF: Production Possibility Frontier

- Feasible: Output combinations that are possible to produce given available resources.
- Efficient: The only way to get more of one good is to get less of the other.
- PPF: All output combinations where inputs are used efficiently to produce output

Individual PPFs

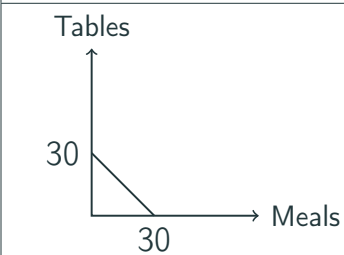
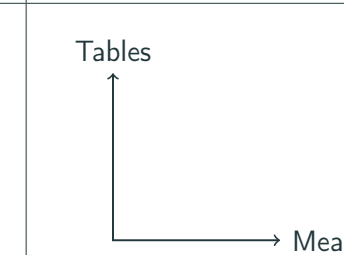
PPF: Production Possibility Frontier

ANN 3 tables/hr or 3 meals/hr 10 hours	BILL 1 tables/hr or 2 meal/hr 10 hours
	

What would they each produce if they lived by themselves?

Increasing Resources

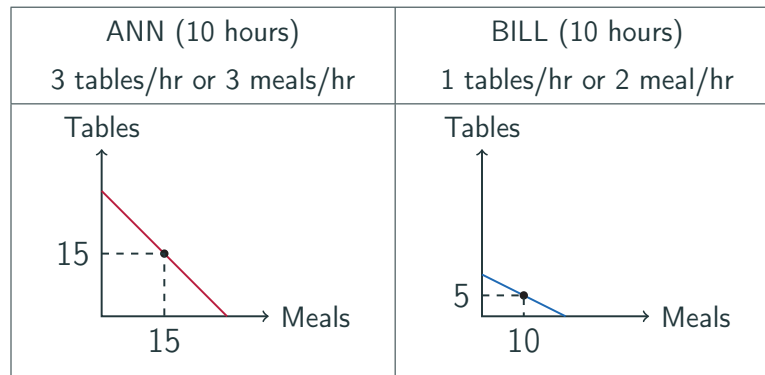
PPF: Production Possibility Frontier

ANN 3 tables/hr or 3 meals/hr 10 hours	ANN 3 tables/hr or 3 meals/hr 20 hours
	

Trade and Individual PPFs

PPF: Production Possibility Frontier

Bill gives Ann 3 meals in return for 2 tables.



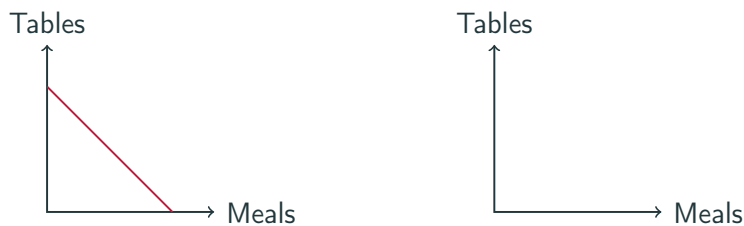
Trade makes both better off.

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Shape of the PPF

PPF: Production Possibility Frontier



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Shape of the PPF

PPF: Production Possibility Frontier; OC: Opp. Cost

- Slope of the PPF is the OC at that point.
- The same constant OC at every quantity gives us a linear PPF
- Changing OC with the quantity produced changes the shape.
- What could give us an increasing OC of tables as more tables are produced?
 - individuals? economy?

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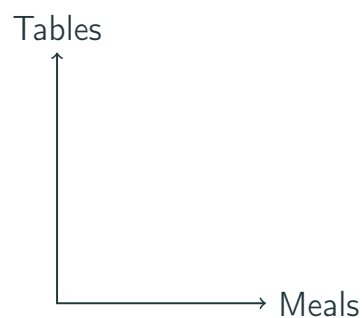
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Joint Production Possibility Frontier (PPF)

- Ann and Bill pool resources and act as one economy.
- Start production by using person with lowest opportunity cost

Joint PPF

PPF: Production Possibility Frontier



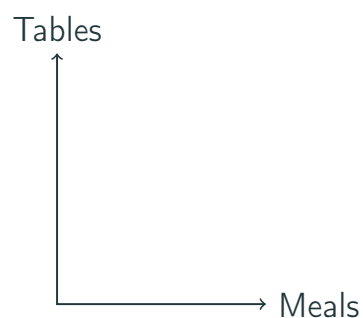
	OC_{TABLE}	OC_{MEAL}
Ann	1 meal	1 table
Bill	2 meals	0.5 tables
10 hours each		

Joint PPF

PPF: Production Possibility Frontier

Ann: 3 tables an hour or 3 meals per hour.

Bill: 1 table an hour or 2 meals per hour.



	OC_{TABLE}	OC_{MEAL}
Ann	1 meal	1 table
Bill	2 meals	0.5 tables
10 hours each		

Trade on the Joint PPF

PPF: Production Possibility Frontier, OC: Opportunity Cost

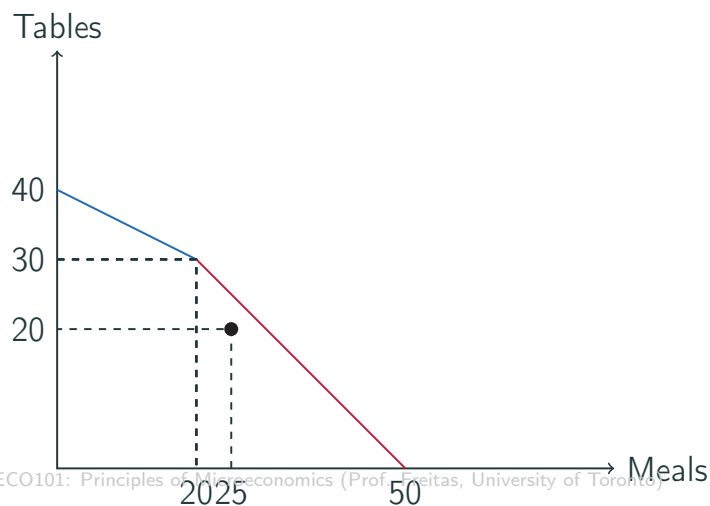
- Before Trade:

	Ann	Bill	Total
Tables	15	5	20
Meals	15	10	25

- Gains from Trade: Possible to produce more output with the same inputs by moving production from high OC to low OC produce (shaded area on graph)

Trade on the Joint PPF

PPF: Production Possibility Frontier



Summary

OC: Opportunity Cost

- Comparative advantage differences $\Rightarrow$ gains from trade.
- If society is producing a good efficiently, it should
 - From among the people able to switch resources into producing that good, choose to person with the lowest OC of production.
 - Put them into production.
 - Their OC is the society wide OC when producing efficiently.